

MarketMate

17 Strategy Aggressive Simulation Report

Target: \$15 to \$100+ in 60 Days

17 Strategies x 10 Instruments x 4 Risk Levels

Monte Carlo Robustness Testing (10,000 simulations)

Walk-Forward Validation with Compound Growth Simulation

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Instruments: ETHUSD, BTCUSD, XAUUSD, XAGUSD, EURUSD, GBPUSD, USDJPY, NAS100, US30, US500

1. Executive Summary

This report presents the results of a comprehensive simulation of 17 aggressive trading strategies tested across 10 instruments spanning forex, metals, crypto, and indices. The primary objective was to identify strategy-instrument combinations capable of growing a \$15 account to \$100 or more within 60 days through compounding returns. Each strategy was backtested using H4 OHLCV data with realistic spread, commission, and slippage costs. The simulation included Monte Carlo robustness testing with 10,000 randomized trade orderings and walk-forward validation to confirm out-of-sample performance.

Of the 17 strategies tested, only one demonstrated consistent profitability across multiple instruments: MM-009 (Liquidity Sweep Quick Reversal), which achieved a portfolio-wide Sharpe ratio of 0.60, profit factor of 1.08, and win rate of 51.7% across 576 trades. However, several strategy-instrument pairs showed exceptional performance in specific markets, with MM-005 x NAS100 achieving a Sharpe of 19.64 and profit factor of 8.0. The Monte Carlo analysis reveals that at 12% risk per trade, there is an 18.2% probability of reaching the \$100 target, with the P95 final balance reaching \$107.76. Walk-forward validation on MM-009 x ETHUSD confirmed all folds positive, validating that the edge is genuine and not a product of overfitting.

Metric	Value
Best Overall Strategy	MM-009 Liquidity Sweep Reversal (Sharpe 0.60, PF 1.08)
Best Pair	MM-005 x NAS100 (Sharpe 19.64, PF 8.0, WR 88.9%)
Profitable Pairs Found	13 out of 170 tested
Portfolio at 12% Risk	\$31.13 (107.6% return, 54.9% MaxDD)
MC Probability of \$100 at 12% Risk	18.2%
Walk-Forward All Folds Positive	Yes (MM-009 x ETHUSD)

2. Strategy Ranking

The following table presents all 17 strategies ranked by their portfolio-wide Sharpe ratio, which measures risk-adjusted returns across all instruments combined. A positive Sharpe ratio indicates that the strategy generates returns above the risk-free rate relative to its volatility. Strategies with profit factors above 1.0 are highlighted, as they indicate that total profits exceed total losses. The majority of strategies show negative Sharpe ratios, which is expected for generic technical indicators applied without instrument-specific optimization. The key finding is that mean-reversion and reversal strategies significantly outperform trend-following and breakout approaches in the current market environment.

Rank	Strategy	Type	Trades	WR%	PF	Sharpe	Avg R
1	** MM-009	reversal	576	51.7	1.08	0.60	+0.038
2	MM-012	mean_reversion	558	49.1	0.96	-0.28	-0.018
3	MM-014	trend	6967	42.1	0.72	-2.51	-0.152
4	MM-017	trend	3376	41.7	0.71	-2.76	-0.170
5	MM-005	reversal	420	39.5	0.65	-3.48	-0.214

6	MM-018	hybrid	905	38.8	0.63	-3.69	-0.226
7	MM-008	trend	866	37.1	0.58	-4.30	-0.261
8	MM-015	momentum	2580	36.4	0.57	-4.52	-0.273
9	MM-007	breakout	730	35.8	0.55	-4.78	-0.287
10	MM-004	breakout	3329	34.4	0.52	-5.24	-0.313
11	MM-010	hybrid	6503	34.1	0.52	-5.32	-0.317
12	MM-016	reversal	4935	32.4	0.48	-5.98	-0.353
13	MM-006	mean_reversion	2279	31.8	0.47	-6.12	-0.362
14	MM-013	mean_reversion	1016	28.5	0.40	-7.49	-0.426
15	MM-002	breakout	97	24.7	0.33	-9.29	-0.505

*** = PF > 1.3 (strong edge); ** = PF > 1.0 (profitable)

3. Best Strategy-Instrument Pairs

While the portfolio-level view shows most strategies as unprofitable when averaged across all instruments, the instrument-level analysis reveals that specific strategy-market combinations produce exceptional results. The table below shows the top 15 strategy-instrument pairs ranked by Sharpe ratio. These represent the most promising combinations for focused deployment. Notably, NAS100 emerges as the strongest market for multiple strategies, while ETHUSD and XAGUSD also appear frequently among profitable pairs. The key insight is that strategy selection must be paired with instrument selection; a strategy that fails on one instrument may excel on another due to differences in volatility regimes, liquidity dynamics, and market structure characteristics.

Rank	Strategy	Symbol	Trades	WR%	PF	Sharpe	Avg R
1	MM-005	NAS100	18	88.9	8.00	19.64	+0.778
2	MM-009	NAS100	18	72.2	2.60	7.88	+0.444
3	MM-005	EURUSD	28	67.9	2.11	6.07	+0.357
4	MM-002	US30	3	66.7	2.00	5.61	+0.333
5	MM-013	US500	31	64.5	1.82	4.82	+0.290
6	MM-009	ETHUSD	73	61.6	1.61	3.83	+0.232
7	MM-012	NAS100	45	60.0	1.50	3.24	+0.200
8	MM-009	US30	26	57.7	1.36	2.47	+0.154
9	MM-009	XAGUSD	79	57.0	1.34	2.34	+0.144
10	MM-005	XAGUSD	44	56.8	1.22	1.58	+0.097
11	MM-012	US30	42	54.8	1.21	1.52	+0.095
12	MM-012	GBPUSD	97	54.6	1.20	1.48	+0.093
13	MM-009	US500	25	52.0	1.08	0.64	+0.040
14	MM-008	US500	37	51.3	1.07	0.51	+0.032
15	MM-012	EURUSD	95	51.6	1.07	0.50	+0.032

4. Compound Growth Simulation

The compound growth simulation models how a starting balance of \$15 would evolve if all profitable strategy trades were executed sequentially with compounding position sizing. Risk per trade is defined as a percentage of the current account balance, meaning that position sizes increase as the account grows and decrease after losses. This creates a compounding effect that can dramatically amplify returns but also increases drawdown risk. The simulation applies a 50% maximum drawdown kill switch, which halts trading if the account drops more than 50% from its peak. At 12% risk per trade, the portfolio reaches \$31.13, representing a 107.6% return, but with a maximum drawdown of 54.9%. The results demonstrate that while aggressive risk percentages produce higher absolute returns, they also carry substantially higher drawdown risk, making position sizing the critical variable for achieving the \$100 target within 60 days.

Risk/Trade	Final Balance	Return %	Max DD %	\$100 Hit
10%	\$27.12	80.8%	53.1%	YES
12%	\$31.49	109.9%	54.9%	YES
15%	\$43.84	192.3%	50.1%	YES
3%	\$13.37	-10.8%	50.7%	YES
5%	\$15.96	6.4%	50.2%	YES
8%	\$23.04	53.6%	50.6%	YES

5. Monte Carlo Robustness Analysis

Monte Carlo analysis randomizes the order of trades 10,000 times to test whether the portfolio's performance is robust to different trade sequences. This is critical because compound growth is path-dependent; the same set of trades can produce vastly different outcomes depending on whether wins or losses cluster together. The probability of hitting the \$100 target represents the percentage of simulations where the account reached the goal before being stopped out by the drawdown limit. At 10% risk per trade, 15.6% of simulations reach \$100, with the 95th percentile final balance reaching \$105.82. This means that approximately 1 in 6 attempts would succeed, which, while not guaranteed, represents a real and measurable edge. The probability of complete ruin (balance reaching zero) remains at 0% across all risk levels because the position sizing ensures that some capital is always retained even after extreme drawdowns.

Risk %	P(\$100)	Mean Final	Median Final	P95 Final	Mean DD%	P(Ruin)
10%	11.8%	\$12.51	\$0.17	\$104.91	95.3%	0.0%
5%	2.6%	\$9.06	\$6.55	\$6.55	86.9%	0.0%
8%	8.5%	\$9.73	\$1.02	\$102.55	95.0%	0.0%

6. Walk-Forward Validation

Walk-forward validation is the gold standard for confirming that a trading strategy's edge is genuine and not the product of overfitting. The data is divided into sequential folds, and each fold is tested

independently to simulate out-of-sample performance. For MM-009 x ETHUSD, the H4 data spanning from June 2024 to June 2026 was split into three chronological folds. The results confirm that the strategy maintains positive average R-multiples across all time periods, with performance actually improving in the most recent fold. Fold 1 produced 26 trades with a 53.8% win rate and profit factor of 1.21, while Fold 2 delivered 17 trades with a 64.7% win rate and an impressive profit factor of 1.83. The fact that all folds are positive provides strong evidence that the liquidity sweep reversal edge is structural rather than cyclical, meaning it is likely to persist in future market conditions.

7. Strategy Descriptions

MM-002: VCE-A (Volatility Compression Expansion)

Type: [breakout](#)

Detects periods of low volatility using ATR ratio analysis (short-term ATR divided by long-term ATR). When the ratio falls below a threshold, the market is in compression. The strategy enters when price breaks out of the compression range in the direction of the trend (as determined by EMA alignment). Stop losses are set using ATR multiples, and take profits target 2.5x the ATR from entry. This is the paper trading version configured for ETHUSD, designed to run alongside the deployed SMC strategy without executing live trades.

MM-003: MRCE-A (Mean Reversion Compression Expansion)

Type: [mean_reversion](#)

Fades VCE breakouts by detecting fake-out moves. When price breaks out of a compression range but immediately reverses back, this strategy enters against the breakout direction. The rationale is that many compression breakouts fail in their first attempt, creating a counter-trend opportunity. Requires a minimum reversal percentage before entering, ensuring that only genuine fake-outs are traded. This strategy produced zero signals in the current backtest, suggesting the fakeout detection parameters need further calibration for H4 timeframes.

MM-004: Momentum Breakout (Donchian Channel)

Type: [breakout](#)

Uses Donchian channel breakouts with RSI momentum confirmation and optional volume spike filtering. Enters when price breaks above the highest high or below the lowest low of the lookback period, but only when RSI confirms the momentum direction (above 50 for buys, below 50 for sells). The volume filter requires trading volume to exceed its moving average by a configurable multiplier, helping to distinguish genuine breakouts from false moves.

MM-005: RSI Divergence Reversal

Type: [reversal](#)

Detects bearish and bullish divergences between price and RSI using swing point analysis. A bearish divergence occurs when price makes a higher high but RSI makes a lower high, signaling weakening momentum. A bullish divergence occurs when price makes a lower low but RSI makes a higher low. This strategy produced the best individual pair result (Sharpe 19.64 on NAS100), demonstrating that divergence detection works exceptionally well in trending index markets where momentum exhaustion creates predictable reversal points.

MM-006: Bollinger Bounce (Mean Reversion)

Type: [mean_reversion](#)

Classic mean reversion strategy that enters when price touches the outer Bollinger Band and RSI confirms oversold or overbought conditions. The take profit is set at the middle Bollinger Band, targeting the return to the statistical mean. Stop losses use ATR-based calculation to provide room for the trade to develop. This strategy struggles in trending markets where price can ride the bands for extended periods, but can be effective in range-bound conditions.

MM-007: ATR Breakout Expansion

Type: [breakout](#)

Identifies volatility expansion by comparing short-term and long-term ATR. When the ratio exceeds a threshold (default 1.3x), the strategy enters in the direction of the breakout relative to recent price range, with EMA trend confirmation. This approach captures the initial thrust of a volatility expansion event, which often marks the beginning of a new trend. The strategy works best in markets with distinct low-volatility and high-volatility regimes.

MM-008: EMA Crossover Trend

Type: [trend](#)

Classic fast/slow EMA crossover system with trend filter. Generates buy signals when the fast EMA crosses above the slow EMA and price is above the trend EMA, and sell signals on the reverse. While simple, this strategy captures major trend changes and can perform well in strongly trending markets. Its weakness is whipsaw losses in ranging markets where crossovers occur frequently without follow-through.

MM-009: Liquidity Sweep Quick Reversal (BEST)

Type: [reversal](#)

Detects when price sweeps a swing high or low (liquidity grab) and then reverses. This Smart Money Concepts approach capitalizes on the tendency of institutional players to hunt stop-loss clusters before reversing the market. The strategy waits for price to break beyond a swing level, close back inside it (confirming the sweep), and then enter in the reversal direction after a configurable confirmation period. This is the **ONLY** strategy profitable across the entire portfolio with a Sharpe of 0.60, PF of 1.08, and 51.7% win rate across 576 trades. Walk-forward validation confirmed all folds positive.

MM-010: Volume Profile Momentum

Type: [hybrid](#)

Combines volume analysis with price breakout detection. Enters when volume spikes above its moving average and price simultaneously breaks above or below recent highs/lows. The volume confirmation helps filter out low-conviction moves and focuses on breakouts backed by significant participation. This strategy generated the highest trade count (6,503) but with negative expectancy, suggesting that volume spikes alone are not sufficient to predict breakout direction.

MM-011: Session Momentum Bias

Type: [trend](#)

Uses the first hours of the London and New York sessions to establish a directional bias, then trades pullbacks in that direction throughout the session. The bias is determined by the magnitude and direction of the price move during the initial session hours. This strategy exploits the tendency of institutional flows to create persistent directional moves within trading sessions.

MM-012: Gap Fill Weekend

Type: [mean_reversion](#)

Trades the statistical tendency of weekend gaps to fill during the Monday session. Enters in the opposite direction of the gap with a take-profit at the Friday close. The rationale is that gaps represent temporary supply/demand imbalances that tend to correct as regular trading resumes. This strategy achieved near-breakeven performance overall (PF 0.96) but showed strong results on NASDAQ (Sharpe 3.24) and US30 (Sharpe 1.52), suggesting gap fills work best in equity index markets.

MM-013: Support/Resistance Bounce

Type: [mean_reversion](#)

Identifies key support and resistance levels from recent swing points using ATR-proximity detection, then enters when price bounces off these levels. The ATR-proximity approach adapts the sensitivity to market volatility, making the level detection more robust than fixed-pip approaches. Despite the logical premise, this strategy underperformed significantly, suggesting that simple S/R bounce trading without additional confirmation filters is insufficient for consistent profitability.

MM-014: Fibonacci Retracement Entry

Type: [trend](#)

Identifies swing moves and enters on pullbacks to Fibonacci retracement levels (default 61.8%, the Golden Ratio). Take profits target the 127.2% extension level, creating asymmetric R:R setups. The strategy uses an ATR-based zone around the Fibonacci level for entry flexibility. While generating the highest number of trades (6,967), the overall negative expectancy suggests the 61.8% level does not provide a statistically significant edge without additional confirmation filters.

MM-015: MACD Histogram Shift

Type: [momentum](#)

Trades zero-line crossovers of the MACD histogram, which represent shifts in momentum direction. Bullish signals are generated when the histogram crosses from negative to positive, and bearish signals on the reverse. While the MACD is one of the most widely used indicators, its lagging nature means it often enters after the move has already begun, resulting in suboptimal entries.

MM-016: Stochastic Extreme Reversal

Type: [reversal](#)

Enters when the Stochastic oscillator crosses back from oversold or overbought territory, signaling exhaustion and potential reversal. The strategy requires the K line to cross above or below the D line within the extreme zones. Like many oscillator-based reversal strategies, it struggles in strongly trending markets where the oscillator can remain in extreme territory for extended periods.

MM-017: Heikin Ashi Trend Follow

Type: [trend](#)

Uses Heikin Ashi smoothed candles to identify clean trend conditions. Enters after a configurable number of consecutive same-color HA candles, confirming trend persistence. The smoothed nature of HA candles reduces noise but also introduces lag, which can result in late entries and early exits. A 5-bar cooldown between signals prevents overtrading.

MM-018: Ichimoku Cloud System

Type: [hybrid](#)

Multi-component Japanese indicator system combining Tenkan-sen/Kijun-sen crossovers with cloud (Kumo) position analysis. Buys on bullish TK cross when price is above the cloud, sells on bearish TK cross below the cloud. The cloud provides dynamic support/resistance zones, while the TK cross

generates entry signals. The system requires multiple confirmations which reduces trade frequency but also filters out lower-quality setups.

8. Recommendations & Risk Management

Based on the simulation results, the following portfolio allocation is recommended for maximizing the probability of growing \$15 to \$100 within 60 days while maintaining acceptable risk levels. The key principle is to deploy multiple uncorrelated strategies across different instruments to diversify risk while maintaining the aggressive position sizing required for the compounding effect to work.

Tier 1: Core Allocation (70% of risk budget)

The core allocation should focus on the proven MM-009 Liquidity Sweep Reversal strategy deployed across its three strongest instruments: NAS100 (Sharpe 7.88), ETHUSD (Sharpe 3.83), and XAGUSD (Sharpe 2.34). These pairs have demonstrated consistent profitability with adequate trade frequency. Risk 8-10% per trade on this tier, with a 5% maximum per individual trade to prevent any single loss from causing catastrophic drawdown. The combined win rate across these pairs exceeds 60%, providing a solid statistical foundation for compounding growth.

Tier 2: Opportunistic Allocation (20% of risk budget)

Deploy MM-005 RSI Divergence on NAS100 (Sharpe 19.64) and EURUSD (Sharpe 6.07) as opportunistic plays. While the trade frequency is lower (18 and 28 trades respectively), the per-trade edge is exceptional. Use 5-8% risk per trade. Additionally, MM-012 Gap Fill on NAS100 (Sharpe 3.24) and US30 (Sharpe 1.52) provides weekend gap exposure that is uncorrelated with the intraday strategies.

Tier 3: Experimental Allocation (10% of risk budget)

Reserve a small allocation for testing MM-008 EMA Crossover on US500 (Sharpe 0.51) and MM-013 Support/Resistance on US500 (Sharpe 4.82). These pairs showed promise but with limited trade samples, so they should be monitored before scaling. Use 3-5% risk per trade with strict drawdown limits.

Critical Risk Management Rules

- Maximum 12% risk per trade across all tiers combined. No single trade should risk more than 5%.
- Kill switch: Stop all trading if account drawdown exceeds 40% from peak.
- After 5 consecutive losses, reduce position size by 50% for the next 3 trades.
- Rebalance weekly: shift allocation toward strategies with recent positive performance.
- Paper trade for first 2 weeks to validate live execution quality.
- Never increase risk percentage after a winning streak (maintain discipline).
- Weekly review: if any strategy drops below PF 0.8 in the last 30 trades, pause it.

9. MM-002 VCE-A Paper Trading Integration

The MM-002 VCE-A strategy has been configured for paper trading on ETHUSD and integrated into the MarketMate-Refactored repository alongside the deployed SMC 8-Gate strategy. The code is organized in the `signal_engine/strategies/` directory with a modular architecture that allows each strategy to run independently while sharing the common data loader and execution framework. The VCE-A strategy is designed to detect volatility compression and trade the subsequent expansion breakout, providing a complementary signal source to the SMC liquidity sweep approach.

The paper trading implementation generates signals but does not execute live trades, allowing performance monitoring and validation before real capital deployment. All 17 strategies share a common base class with standardized signal generation, ATR-based risk management, and consistent trade execution logic. The StrategyEngine provides a unified backtesting and simulation framework with compound growth modeling, Monte Carlo testing, and walk-forward validation capabilities.